

Bayesian Estimation of DSGE Models

Priors, posterior inference and model diagnostics

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SESSION OBJECTIVE

- ▶ Combine economic prior information with the likelihood.
- ▶ Understand posterior simulation and convergence diagnostics.
- ▶ Produce a reproducible Bayesian DSGE analysis in Dynare.

Handout: Session 4 **Code:** `code/session4/`

PLAN

- 1 Bayesian updating
- 2 Posterior computation
- 3 Dynare implementation
- 4 Course synthesis

BAYES' RULE FOR MODEL PARAMETERS

$$p(\theta \mid y) = \frac{p(y \mid \theta) p(\theta)}{p(y)} \propto \underbrace{p(y \mid \theta)}_{\text{likelihood}} \times \underbrace{p(\theta)}_{\text{prior}}.$$

- ▶ The prior states information and economically plausible support before observing the sample.
- ▶ The posterior quantifies what the model and data imply jointly.
- ▶ Transparency requires reporting both prior and posterior.

WHAT MAKES A USEFUL PRIOR?

Good practice

- ▶ respect parameter support
- ▶ state mean and dispersion
- ▶ relate choice to economics or evidence
- ▶ test sensitivity when relevant

Avoid

- ▶ hidden tight priors
- ▶ priors chosen after seeing results
- ▶ interpreting a posterior without its prior

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FROM POSTERIOR MODE TO SIMULATION



The sampler approximates posterior expectations with draws, not a single point estimate.

MCMC DIAGNOSTICS ANSWER CONCRETE QUESTIONS

- ▶ **Mixing:** do chains explore the posterior efficiently?
- ▶ **Convergence:** do independent chains give compatible summaries?
- ▶ **Effective sample size:** how much independent information remains after autocorrelation?
- ▶ **Prior–posterior comparison:** did the data move beliefs?

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ESTIMATION BLOCK: COMMUNICATE ASSUMPTIONS

```
estimated_params;  
  rho, beta_pdf, 0.80, 0.10;  
  stderr e, inv_gamma_pdf, 0.01, 2;  
end;  
estimation(datafile=mydata, mode_compute=6,  
           mh_replic=20000, mh_nblocks=2);
```

Read the code as a research protocol: observables, priors, sample, algorithm and output choices must be explicit.

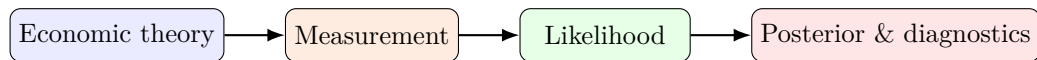
A DEFENSIBLE EMPIRICAL WORKFLOW

1. Download and document data; transform it to model units.
2. Verify the deterministic steady state and stochastic solution.
3. Inspect prior predictive implications and estimate.
4. Diagnose convergence and compare prior/posterior distributions.
5. Interpret impulse responses and uncertainty economically.

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FROM THEORY TO CREDIBLE INFERENCE



A good report makes every link in this chain inspectable and reproducible.

ASSESSMENT REMINDER

- ▶ Choose one published model and write the report in English.
- ▶ Submit a LaTeX-produced PDF and complete reproducible code.
- ▶ Explain data, transformations, estimates and diagnostics.
- ▶ Do not include caches or unnecessary outputs.

Submission deadline: 15 January 2027